Technical and Business Writing: An Introduction

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Objectives

Upon completion of this ppt, students will be able to:

1. Define technical writing.
2. Summarize the six characteristics of technical writing.
3. Explain basic standards of good technical writing.
4. Understand the need for technical and business writing in Software Engineering

COMMUNICATION

First, a little about communication...

What is communication?

The word communication is derived from the Latin word ‘communicare’ which means to make common, to transmit, or to impart the ideas, knowledge, feelings, emotions and gestures.

The Process of Communication

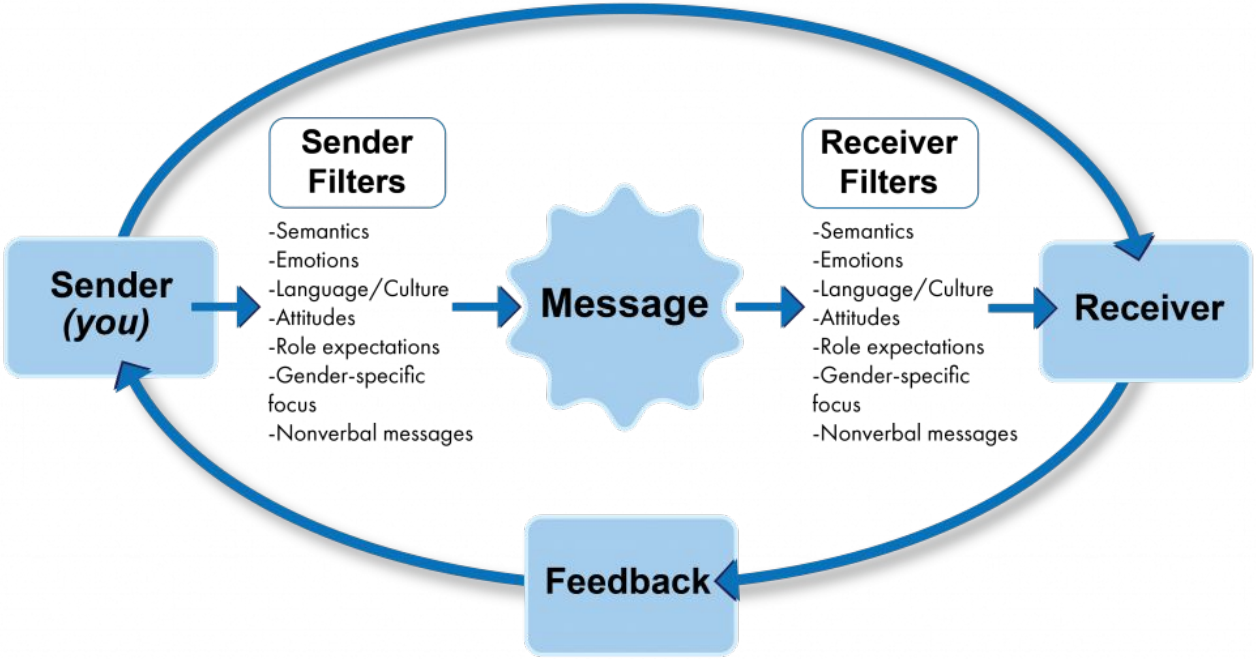
SENDER - transmits a MESSAGE
(ENCODING - The sender must choose certain words or nonverbal methods to send an intentional message)

CHANNEL/MEDIUM - method of message transmission (letter, memo, fax email, etc.)

RECEIVER - interprets/responds
(DECODING - attaching meanings and symbols)

FEEDBACK - response of the receiver to the sender

Communication Process



Communication Contexts

- Physical
  - Influences the content and quality of interaction
- Social
  - Nature of relationship between sender and receiver
- Chronological
  - How time influences interaction
- Cultural
  - Organizational background
  - Ethnic background
  - National background

Principles of Communication

- Communication Is Unavoidable
- Communication Operates on Two Levels
  - Content messages (information about the topic under discussion)
  - Relational messages (signals indicating how they feel about one another)
    - Affinity
    - Control
    - Respect
- Communication Is Irreversible
- Communication Is a Process
- Communication Is Not a Panacea

Communication Networks

- Formal Communication Network
  - Downward communication (superiors to subordinates)
  - Upward communication (Subordinates to superiors)
  - Horizontal/Lateral communication (messages between members of an organization with equal power)
- Informal Communication Network
  - Grapevine
    - Single Strand
    - Gossip
    - Probability
    - Cluster
  - Small talk
  - Networking

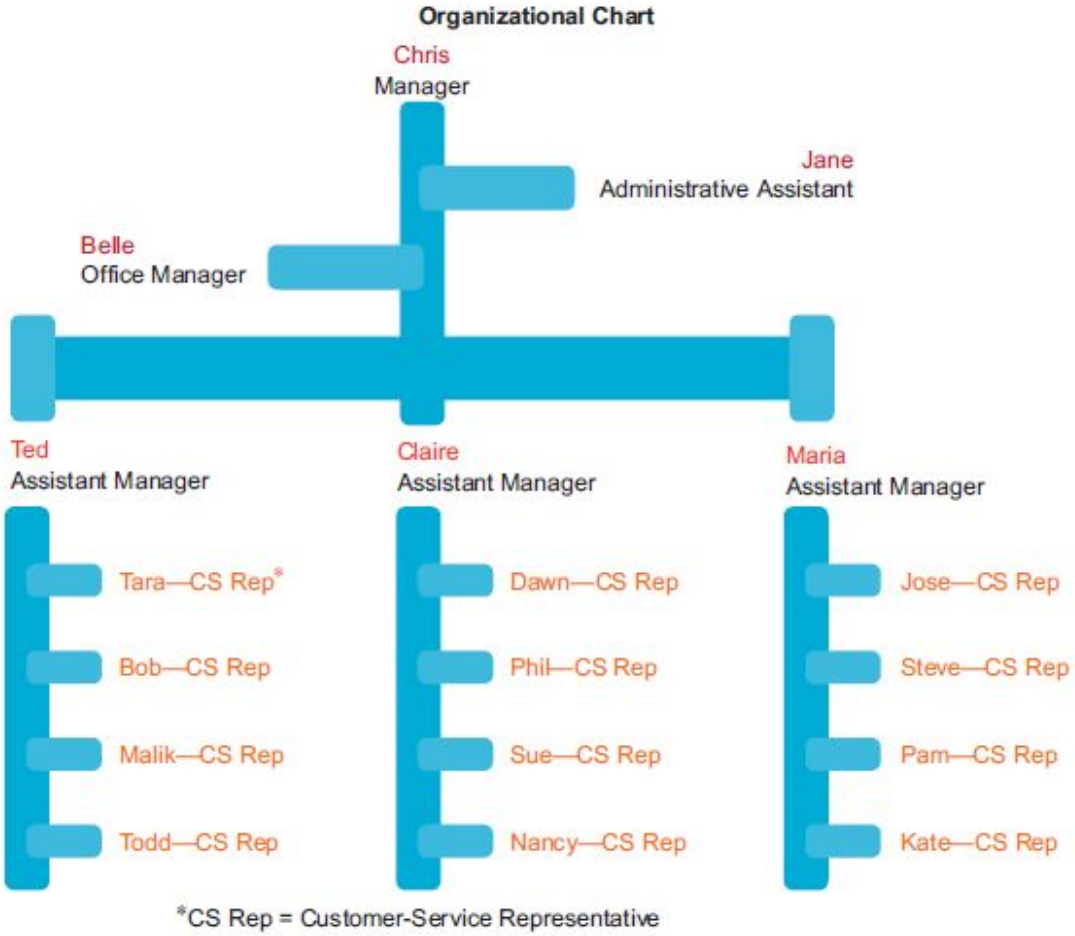


FIGURE 1–2
A Formal Communication Network

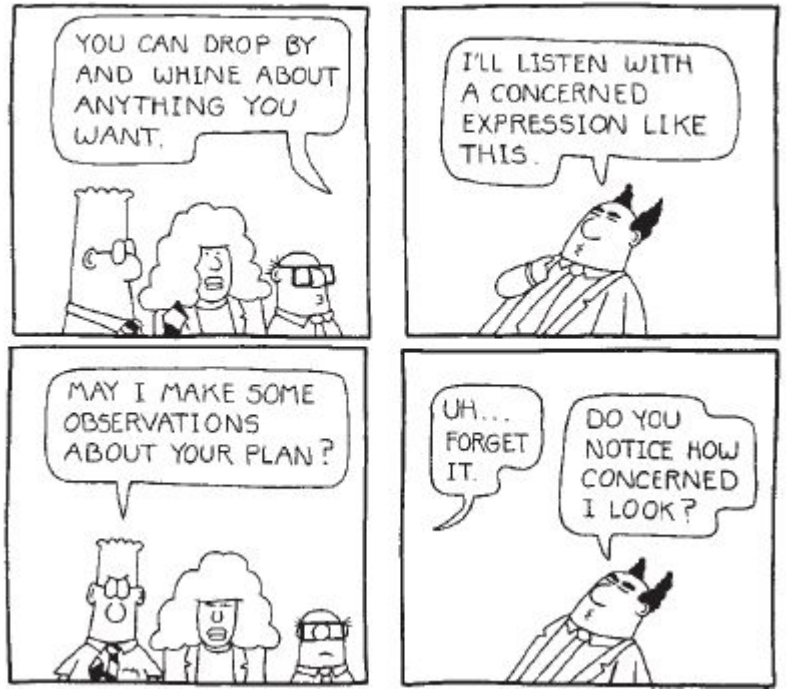
Formal communication: Downward Communication

- Job instructions
- Job rationale
- Procedures and practices
- Feedback
- Indoctrination



Formal communication: Upward Communication

- What subordinates are doing
- Unsolved work problems
- Suggestions for improvement
- How subordinates feel about each other and the job

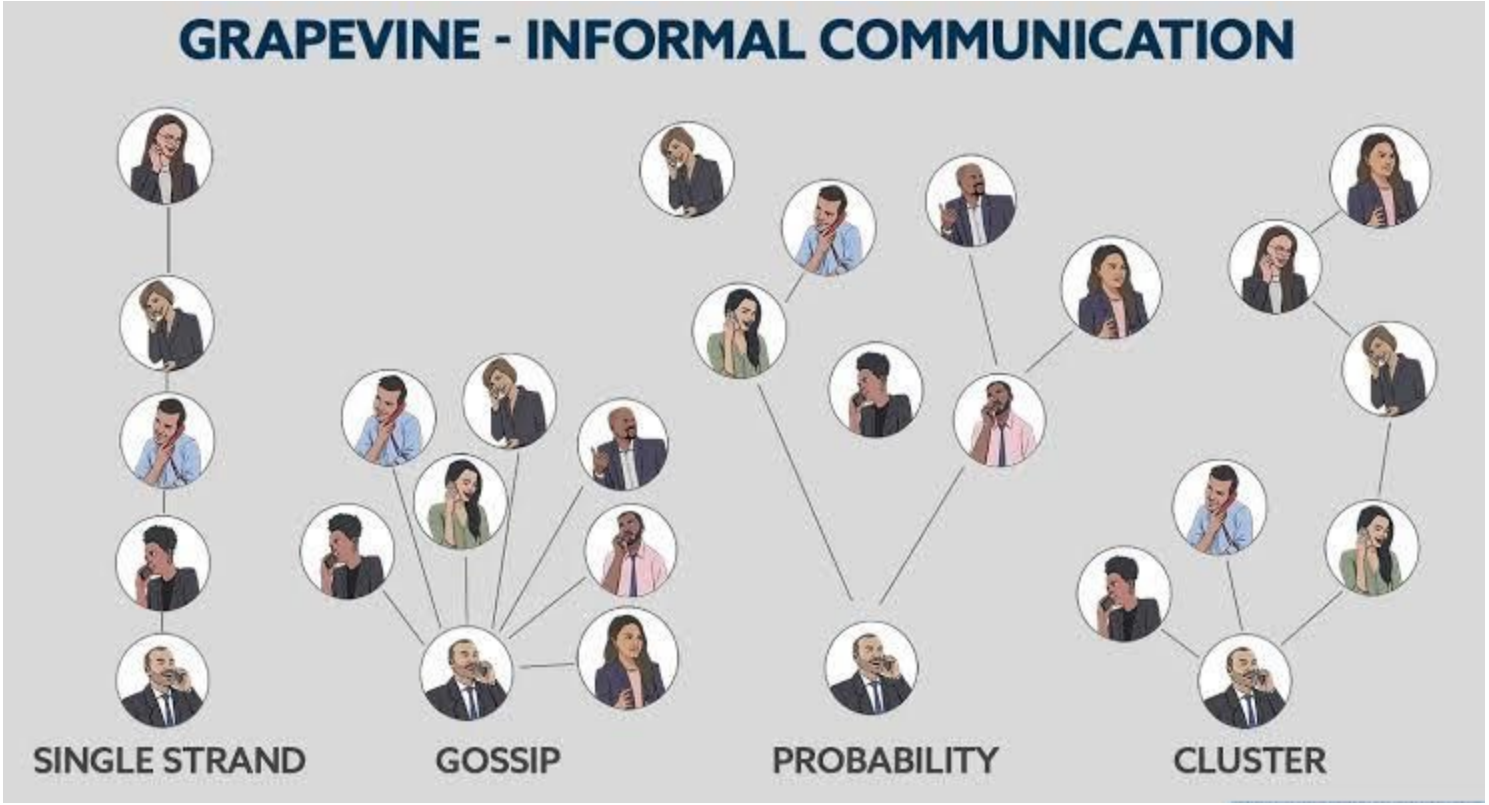


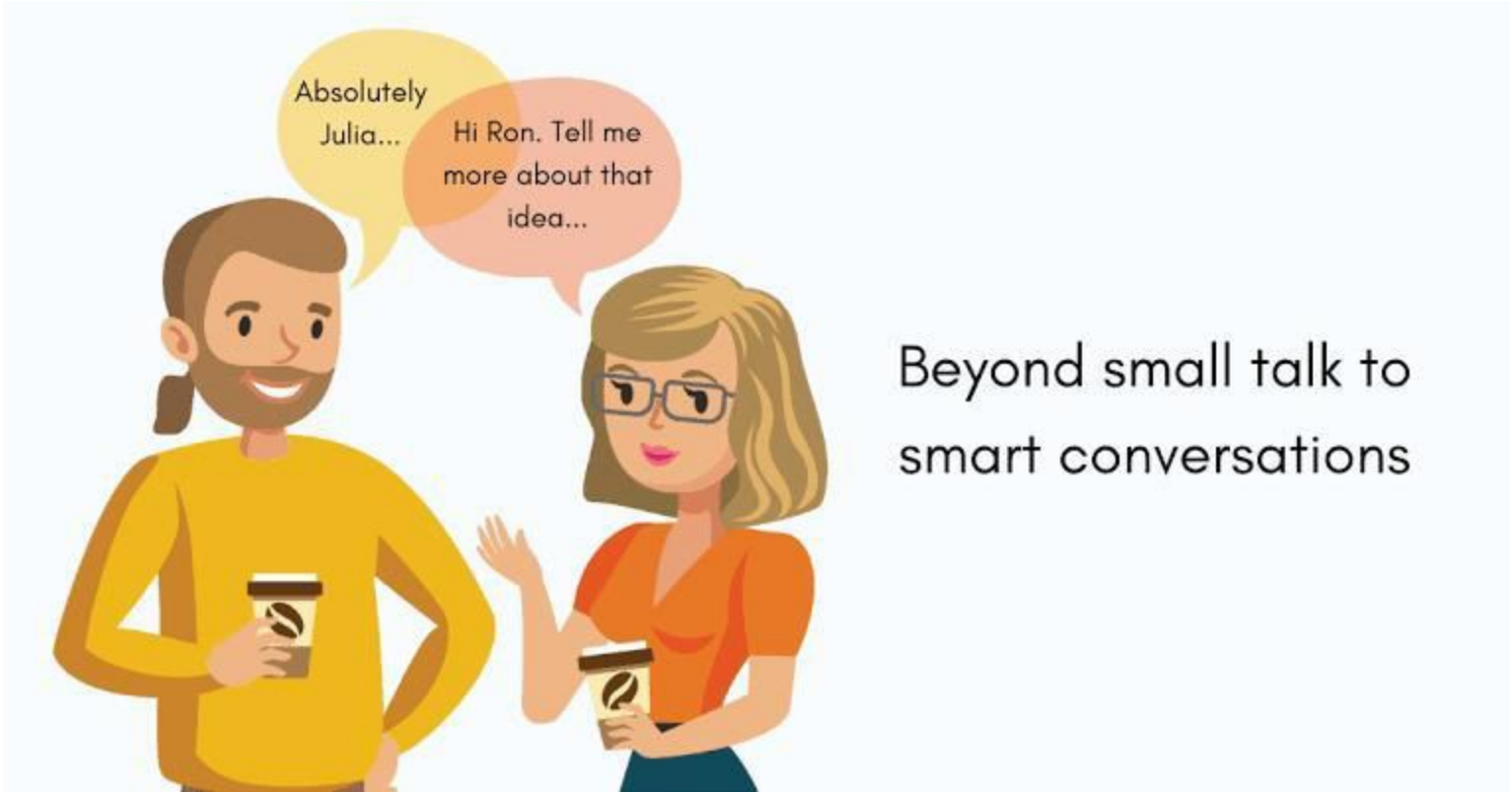
Formal communication: Horizontal Communication

- Task coordination
- Problem solving
- Sharing information
- Conflict resolution
- Building rapport

Informal communication

- *Confirming formal messages:* "The boss is really serious this time about cutting down on overnight travel. I heard him yelling about it when I walked past his office."
- *Expanding on formal messages:* "The invitation to the office party says 'casual dress,' but don't make it too informal."
- *Expediting official messages:* You might learn about openings within an organization from people in your network long before the vacancies are published.
- *Contradicting official messages:* You might learn from a friend in accounting that the deadline for purchases on this year's budget isn't as firm as it sounded in the controller's recent memo.
- *Circumventing formal channels:* Your tennis partner who works in duplicating might sneak in an occasional rush job for you instead of putting it at the end of the line.


GRAPEVINE - INFORMAL COMMUNICATION
SINGLE STRAND
GOSSIP
PROBABILITY
CLUSTER


Absolutely Julia...
Hi Ron. Tell me more about that idea...
Beyond small talk to smart conversations



Networking

Considerations in Choosing a Communication Channel

	Time required for feedback	Richness of information conveyed	Sender s control over how message is composed	Control over receiver s attention	Effectiveness for detailed messages
Face-to-Face	Immediate	High	Moderate	Highest	Weak
Telephone Teleconferencing, and Videoconferencing	Immediate	Moderate	Moderate	Moderate	Weak
Voice Mail	Delayed	Moderate	High	Low	Weak
E-Mail	Varies	Low	High	Low	High
Instant Messaging	Varies	Low	High	Varies	Weak
Text Messaging and Twitter	Can be immediate	Low	High (given briefness of message)	Low	Good for brief messages
Hard Copy (e.g., handwritten or typed message)	Delayed	Low	High	Low	High

So, what's a technical writer?

The short definition of "technical writer" is a person who writes about technical topics. But perhaps a better definition is someone who can explain complicated concepts in clear, easy-to-understand prose.

A technical writer is really a translator. You start with a complicated piece of technology, and your mission is to explain to a nonexpert how to use that technology.

This deceptively simple mission requires more than just writing ability and an understanding of technology—although both of those skills are critical, they aren't enough.

So, what is technical writing?

If you

- noticed signs on your way to work,
- checked the calories on the cereal box,
- emailed your professor to request a recommendation,
- or followed instructions to make a withdrawal from an ATM;

you have been involved with technical, workplace, or professional communication.

WHY THE NEED FOR IT?

Today, writing is a more important skill for professionals than ever before.

A survey by the Workforce Solutions group at St. Louis Community College asserts many employers are concerned at the large number of college graduates applying for jobs who lack communication and interpersonal skills (White, 2013).

Good communication skills: essential for success in the workplace

Within modern organizations, almost every action is documented in writing.

Characteristics of Technical Writing

Focused on audience

Rhetorical, persuasive, purposeful, and problem-oriented

Professional

Design centered

Research and technology oriented

Ethical

What Standards Should I Observe to Make my Writing Successful?

- Honest
- Clear
- Accurate
- Complete
- Attractive
- Correct
- Legible
- Readable

Accessibility in Technical Writing

The most important standard for excellence in technical communication.

The design of your document should be useful, easy to navigate, and with all information easy to locate

Technical and Business Writing in STEM fields

Technical and business writing in software engineering turns complex code and project goals into clear, structured facts for users, developers, and company leaders

- **API Documentation**: Guides for developers on how to use code libraries, tools, and endpoints.
- **User Manuals**: Instructions that help everyday users navigate software interfaces.
- **Design Documents**: Internal blueprints that map out software architecture before coding begins.
- **Release Notes**: Summaries of new features, bug fixes, and updates in a software version.

Business Writing in Software Engineering

Business writing focuses on strategy, value, and professional communication. It helps teams align on money, timelines, and project direction.

- **Project Proposals**: Documents that pitch a new software feature or tool to decision-makers.
- **Statements of Work (SOW)**: Contracts that outline project tasks, deliverables, and schedules.
- **Status Reports**: Updates sent to stakeholders regarding project progress and risks.
- **Internal Memos**: Clear notes used to share policy changes or team shifts.

Key Differences

	Technical writing	Business writing
Purpose	explains *how* a product works;	explains *why* a project makes financial or operational sense
Audience	targets developers or end-users	targets managers, clients, and executives
Tone	uses a strict, literal tone	can be persuasive and strategic

Next week:

Characteristics of Effective Writing

Audience Analysis

Purpose and Context